

6. The table below shows the mean carbon footprint for 1 kWh of electricity generated in different ways.

Type of power station	Carbon footprint (gCO ₂ eq/kWh)
coal	921
oil	750
gas	500
solar	100
biomass	52
nuclear	30
hydroelectric	27
wind	30

Use the information in the table and your knowledge to answer the following questions.

(a) (i) Define the term carbon footprint. [2]

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(ii) A typical 500 MW **coal**-fired power station produces 3.5 billion kWh of electricity per year.
Calculate the carbon footprint of this power station in gCO₂eq/kWh per year.
Give your answer to 2 significant figures.

1 billion kWh = 1 × 10⁹ kWh [4]

carbon footprint = gCO₂eq/kWh



- (iii) The carbon footprint of a nuclear power station is close to zero whilst generating electricity.

Explain why the table shows a carbon footprint of $30 \text{ gCO}_2\text{eq/kWh}$ for power generated by nuclear power. [2]

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- (iv) A biomass power station burns wood chips as its energy source.
State why a biomass generator is considered as being 'carbon neutral'. [1]

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- (b) The Antarctic ice sheet traps methane. It is estimated that the mass of trapped methane is about $2.8 \times 10^{17} \text{ kg}$ which has a carbon dioxide equivalent of $7 \times 10^{18} \text{ kgCO}_2\text{eq}$.

- (i) Explain what is meant by the term global warming potential. [1]

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- (ii) Use the equation:

carbon dioxide equivalent = mass of a gas $\times$ global warming potential of the gas
to calculate the global warming potential of methane. [2]

global warming potential =

END OF PAPER





THE PERIODIC TABLE

Group

1

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3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2																																																													
7 Li Lithium 3				9 Be Beryllium 4				11 B Boron 5				12 C Carbon 6				14 N Nitrogen 7				16 O Oxygen 8				19 F Fluorine 9				20 Ne Neon 10																																											
23 Na Sodium 11				24 Mg Magnesium 12				27 Al Aluminium 13				28 Si Silicon 14				31 P Phosphorus 15				32 S Sulfur 16				35.5 Cl Chlorine 17				40 Ar Argon 18																																											
39 K Potassium 19				40 Ca Calcium 20				45 Sc Scandium 21				48 Ti Titanium 22				51 V Vanadium 23				52 Cr Chromium 24				55 Mn Manganese 25				56 Fe Iron 26				59 Co Cobalt 27				59 Ni Nickel 28				63.5 Cu Copper 29				65 Zn Zinc 30				70 Ga Gallium 31				73 Ge Germanium 32				75 As Arsenic 33				79 Se Selenium 34				80 Br Bromine 35				84 Kr Krypton 36			
86 Rb Rubidium 37				88 Sr Strontium 38				89 Y Yttrium 39				91 Zr Zirconium 40				93 Nb Niobium 41				96 Mo Molybdenum 42				99 Tc Technetium 43				101 Ru Ruthenium 44				103 Rh Rhodium 45				106 Pd Palladium 46				108 Ag Silver 47				112 Cd Cadmium 48				115 In Indium 49				119 Sn Tin 50				122 Sb Antimony 51				128 Te Tellurium 52				127 I Iodine 53				131 Xe Xenon 54			
133 Cs Caesium 55				137 Ba Barium 56				139 La Lanthanum 57				179 Hf Hafnium 72				181 Ta Tantalum 73				184 W Tungsten 74				186 Re Rhenium 75				190 Os Osmium 76				192 Ir Iridium 77				195 Pt Platinum 78				197 Au Gold 79				201 Hg Mercury 80				204 Tl Thallium 81				207 Pb Lead 82				209 Bi Bismuth 83				210 Po Polonium 84				210 At Astatine 85				222 Rn Radon 86			
223 Fr Francium 87				226 Ra Radium 88				227 Ac Actinium 89																																																															
										Key																																																													

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

20

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